Stanford

Weekly update 3 of 5 · week of Nov 23

Adding knowledge: v2 and documents

THIS UPDATE IN THREE NUMBERS

13.8%

run-to-run CV (v1 19.0%): more consistent

15.1%

coverage (v1 19.5%): less accurate

0.61

bias with envelope documents in context (v1 0.72): worse

Stanford
School of Engineering &
Doerr School of Sustainability
Civil & Environmental Engineering

What prompt v2 changed

1 Scope

Written to match the dataset's enclosure definition.

2 Component take-off

Areas → assemblies → intensities → total.

3 High-carbon materials

HFC-blown XPS "ten times or more" the GWP of other insulation per unit R; aluminum curtain wall 100–200 $kgCO_2e/m^2$.

4 No anchoring

Estimating from a "typical building" is ruled out.

5 Renovation rule

Count only the materials of the envelope work.

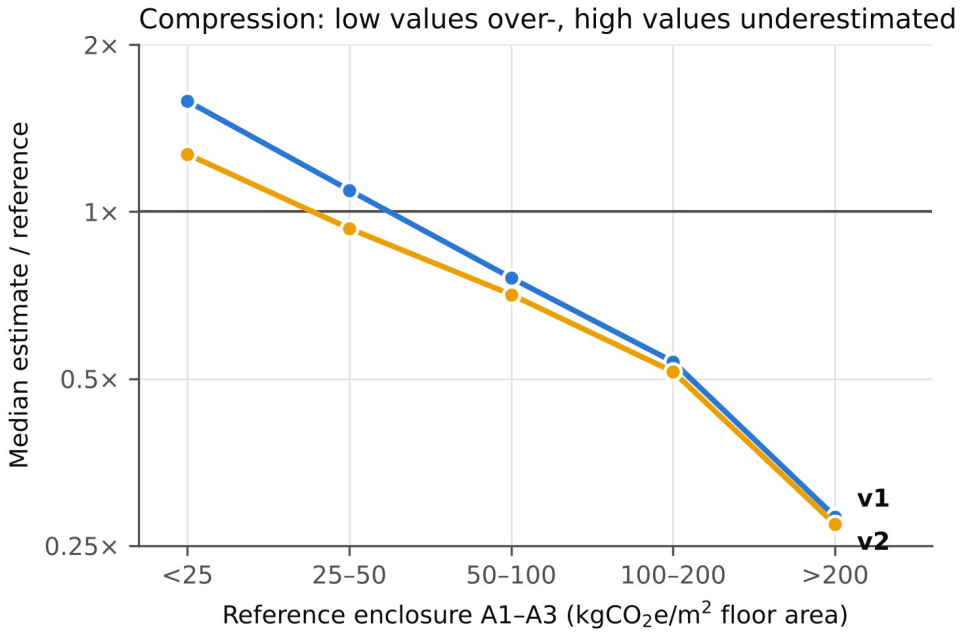
6 BEST value

A best estimate is requested alongside the range.

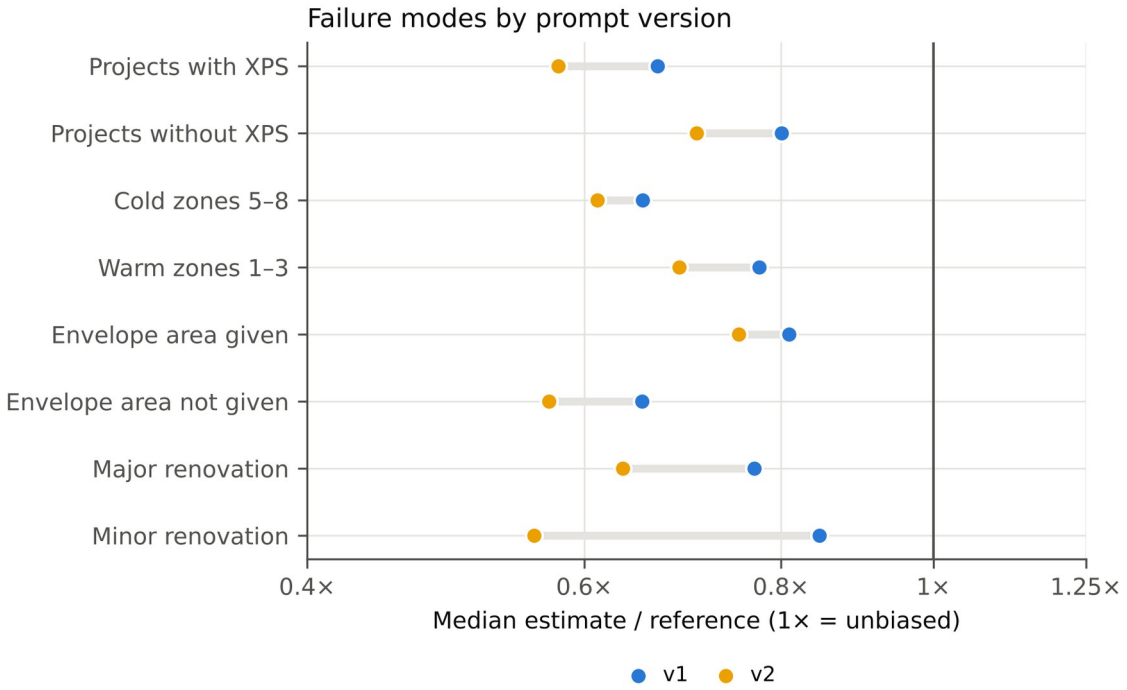
Results: v2 vs v1 (292 buildings, 3 runs each)

Prompt	Coverage	RMSE	Med. APE	Med. |log|	Bias	CV
v1	19.5%	79.7	37.9%	0.456	0.75	19.0%
v2	15.1%	81.0	44.3%	0.563	0.67	13.8%

- v2 errors are **significantly larger** (Wilcoxon $p < 0.0001$).
- Compression eased (slope 0.46 → 0.55), but the **whole curve shifted down**.
- More consistent across runs: CV 19.0% → 13.8%.



Why v2 got worse



1. **The facts were repeated, not applied.** XPS was named 50× more often, but 89% of answers still assumed polyiso roofs; the median roof intensity stayed at 35 $kgCO_2e/m^2$.
2. **The take-offs were too light.** Bottom-up intensities came out at 66.8 per m^2 of envelope, against 86.7 in the reference.
3. **A subtracting rule hurt.** "Count only new materials" dropped minor renovations from 0.85 to 0.56.

Second way to add knowledge: envelope documents in context

Condition (run 0)	Coverage	Med. |log|	Bias	RMSE
v1 (innate)	17.8%	0.519	0.72	81.2
v1 + 4 envelope docs	15.8%	0.609	0.61	85.3

- Docs: RDH 2025 wall assemblies, CLF 2023 material baselines, Arup facades/glass, UBC fenestration (30k characters each); v1 task text unchanged.
- **Worse:** log error +0.090 (Wilcoxon $p < 0.001$); bias 0.72 → 0.61.

Why

- **The documents were read:** 22% of answers cite them, mostly for the low end of a range.
- But they describe **today's low-carbon products** (HFO foams, recent EPDs, modern glazing); the reference LCAs used period tool data.
- XPS mentions fell to **0%**; the XPS-project ratio fell 0.64 → 0.50.

Lesson for prompt design

Qualitative knowledge, as instructions or as documents, gets repeated, not applied. To move the numbers, give numbers, at the step where the model chooses them.

This is the design principle behind **prompt v3**.

Next update

Coming next

- **Prompt v3:** sourced quantitative anchors. A table of external A1–A3 intensities per m² of component, placed at the decision step.
- **Held-out check** on buildings not used for the prompt diagnostics.

Input we'd like

- Sources for intensities: CLF baselines, PCI and Kawneer EPDs, RDH studies. **None from the ground-truth dataset or the CLF WBLCA benchmark** (leakage). Any to add or avoid?